

Introduction

Clinical laboratories in the USA are suffering from staffing shortages. There are physically not enough certified professionals to staff the country. Two main solutions to this problem are to import immigrants to fill job positions, or more automation.

Many lab processes have already been automated. However, microscopy has traditionally been a very hands-on process. The goal of this project is to automate manual blood cell differentials which are traditionally performed manually by a laboratory tech.

Automation of this procedure can allow lab techs to keep current staffing levels while maintaining or increasing the laboratory workload.

Current analyzers exist that identify cell types based on images. However, they are limited by supplier and facilities budget.

Project

The goal of this project is to make an automated model that will identify white blood cells from microscope photos and sort them into 5 or more categories.

The data sets are sourced from Kaggle. Multiple image data sets will be used to create a more robust model.

Data will be cleaned and processed in order to create image models using scikit learn, and neural nets.

The models will be exported and findings presented in a project report and slide deck.